# Sport & Exercise Biomechanics 2

# Sport and Exercise Science

**May, 2026**

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

# Sport & Exercise Biomechanics 2 (A22542)

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| **Short Title:** | Sport & Exe Biomechanics 2 | | |
| **Faculty** | Science and Computing | | |
| **Department** | Sport and Exercise Science | | |
| **Cluster** | Sports Technology | | |
| **Author** | RBOLGER | | |
| **Credits** | 5 | **Level:** | Intermediate |

## Description of Module / Aims

The aim of this module is to develop students understanding of the key concepts in sport and exercise biomechanics and also to develop their practical skills in the collection and analysis of human movement data.

## Programmes

| | | |
|---|---|---|
| `SPOR-0124` | BSc (Hons) in Software Systems Development (`WD_KDEVP_B`) | 4 / 7 / E |
| | BSc (Hons) in Sport and Exercise Science (`WD_T0012_X`) | 3 / 5 / M |

## Indicative Content

- Three dimensional kinematics - linear and angular
- Three dimensional kinetics - linear and angular
- Power, force-velocity relationship
- Quantitative analysis of human movement
- Systematic qualitative analysis of human movement (including deterministic models of performance, prioritisation, cueing etc.)
- Biomechanics of specific skills/sports (e.g. long jump, throwing, kicking etc.)
- Advanced analysis of human movement data (e.g filtering, interpolation, normalisation)
- Practical aspects of biomechanics testing equipment (e.g. motion analysis, EMG, force platforms, GPS etc.)
- Muscle mechanics

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Describe the principle biomechanical concepts underpinning movement in sport.
2. Perform advanced calculations relating to kinematic and kinetic quantities.
3. Compare the practical benefits and limitations of equipment used during biomechanical testing.
4. Organise the collection of biomechanical data of a sports or exercise movement.
5. Complete detailed analyses of human movement.

## Learning and Teaching Methods

- Lectures - incorporating interaction and engagement with prior reading
- Practicals - providing skills for self-directed data collection and project completion
- Tutorials - task-based and problem-based learning

## Assessment Methods

| | Weighing | Outcomes Assessed |
|---|---|---|
| Final Written Examination | 50% | 1,2,3 |
| Continuous Assessment | 50% | 1,2,3,4,5 |
| *In-Class Assessment* | 10% | 1,2,3 |
| *Project* | 40% | 4,5 |

## Assessment Criteria

**Notes:**

- Final Examination - written paper
- Mid-term in-class exam (10%)
- Project - individual scientific project assessing planning, data collection, analysis and report preparation skills

## Learning Modes

| Learning Mode | F/T Hours | P/T Hours |
|---|---|---|
| Lecture | 12 | |
| Tutorial | 12 | |
| Practical | 12 | |
| Independent Learning | 99 | |

## Essential Material

- McGinnis, P.M. _Biomechanics of Sport and Exercise_. 3rd ed. Champaign, IL: Human Kinetics, 2013.

## Supplementary Material

- Hamill, J. and K.M. Knutzen. _Biomechanical Basis of Human Movement_. 4th ed. Philadelphia, PA: Lippincott, Williams and Wilkins, 2015.
- Knudson, D. _Fundamentals of Biomechanics_. USA: Springer, 2007.
- Knudson, D. _Qualitative Diagnosis of Human Movement_. Human Kinetics: 3rd ed, 2013.
- Winter, D.A. _Biomechanics and Motor Control of Human Movement_. 4th ed.. UK: Wiley, 2009.

## Requested Resources

- Room Type: Computer Lab